22 30 00 - Plumbing Equipment

Domestic Water System

Cold water systems will be sized using CPC flush valve curves, and hot water systems using flush tank curves. Equipment branches and main pipe size will be based on flow requirements without diversity. All plumbing valves will be located behind a lockable access panels when they are in a concealed location.

Hose bibs will be provided on the roof of each building. The hose bibs will be spaced 50 ft apart, maximum.

Chrome plated stops with gasket seats will be provided for sinks, lavatories, and wash basins when exposed to public view. A hose bibb with vacuum breaker will be provided under the lavatories in each toilet room. Exposed branch water supply piping in toilet rooms and custodial rooms will be chromium plated.

Water hammer arrestors will be provided in the wall, as required, behind a lockable access panel. Trap primers from lavatory tailpieces or water closet flush-o-meter valves for floor drains will also be provided. Isolation valves and unions at equipment connections will be provided.

The water piping will be designed to provide a minimum residual pressure at the most remote water closet of at least 30 psig. Pressure regulators will be installed to comply with California Plumbing Code.

- The maximum water pipe velocity will be 5 ft/sec. for hot water and 8 ft./sec for cold water.
- The minimum supply pipe size will be ½” for one plumbing fixture with a maximum flow of 1.25 gpm.
- Fixture pipe sizes will be as follows: 1-1/2” for a flush valve water closet, ¾” for a shower, ½” for sink and ¾” for hose bibs.

Domestic water piping within the buildings will be “Viega” pro-press or equal copper tubing and shall conform to ASTM B 75 or ASTM B88, Fittings shall be copper in conformance with ASTM B16.18, ASTM B16.22 or ASTM B16.26. O-rings for copper press fittings shall be EPDM.22 32 00 Domestic Water Filtration Equipment

Building Pure Water Systems

Building (house) pure water systems shall be designed to guarantee a water quality level of at least Lab Grade 3 per National Committee for Clinical Laboratory Standards (NCCLS) at all times. Individual laboratories requiring a guarantee of purer water shall install additional purification equipment in the laboratory as part of the building equipment (reference; Letter of Understanding dated 3-19-90, J. West; Physical Plant to K. McCaffrey; Natural Sciences).

The makeup water shall include pre-treatment consisting of a prefilter, a water softener, an activated carbon absorption filter, and a five (5) micron filter followed by treatment which shall consist of a reverse osmosis (RO) unit, mixed bed deionizer exchange service tanks, an ultra-violet light, and a 0.2

micron filter. Make up water capacity shall be determined in conjunction with sizing of the system storage tank.

The entire piping system shall be continually circulated. Dead legs shall be limited to drops to individual lab outlets. (Confirm maximum dead leg distance for each application) The entire circulating volume shall be exposed to ultra-violet light on each circulation. Provide dual stainless steel circulation pumps to allow for continued circulation in the event of pump failure.

The system shall include a storage tank sized to allow make up to occur over a 12 hour period while accommodating maximum anticipated use with 20 % excess capacity. Determination of maximum system use shall include a diversity factors which shall be determined with the University's Representative for each system.

The system shall include a side stream polishing loop which will continually circulate a portion of the water circulating through the system back through the mixed bed deionizers prior to return to the storage tank.

Provide a back pressure valve on the return piping to the storage tank set to assure adequate pressure at the highest most laboratory faucet.

Provide pressure gauges on both sides of all filters and pumps.

Building (house) pure water systems shall be monitored by the Building Management System to provide alarms to the Physical Plant watch stander at the campus Central Heat Plant. Alarms shall be generated in the case of Water quality (conductivity) out of range, low tank level, and no flow.

Acceptable Piping Materials:
Socket fused polypropylene prepared specifically for deionized service. Pipe shall be sterilized and capped immediately after production. Fittings, valves, and unions shall be sterilized and individually wrapped immediately after production. Continuous trough support system.

Schedule 80 PVC with solvent weld fittings prepared specifically for deionized service. Pipe shall be sterilized and capped immediately after production. Fittings, valves, and unions shall be sterilized and individually wrapped immediately after production. Continuous trough support system. Stainless steel could be used in some applications.

Confirm piping material with the University's representative on a case by case basis.

Valves shall be diaphragm type.

Valving:
Pure water piping systems shall be provided with manual isolation valves at the following locations:

At each floor where branch piping connects to a riser.

On both sides of piping elements which may need to be removed for servicing including filters, pumps, ultraviolet lights, mixed bed deionizers, and RO Units.

At branch connections to mains.
Provide provisions for system drain down to a floor drain including drain down of storage tanks.